Melt Temperature Verification DOE in Plastic Injection Molding

Technical Guide for Engineers, Process Technicians, Tooling Specialists, and Manufacturing Leaders

Cold Runner vs. Hot Runner Methodology, Diagnostics, Corrective Actions, and Control Plan

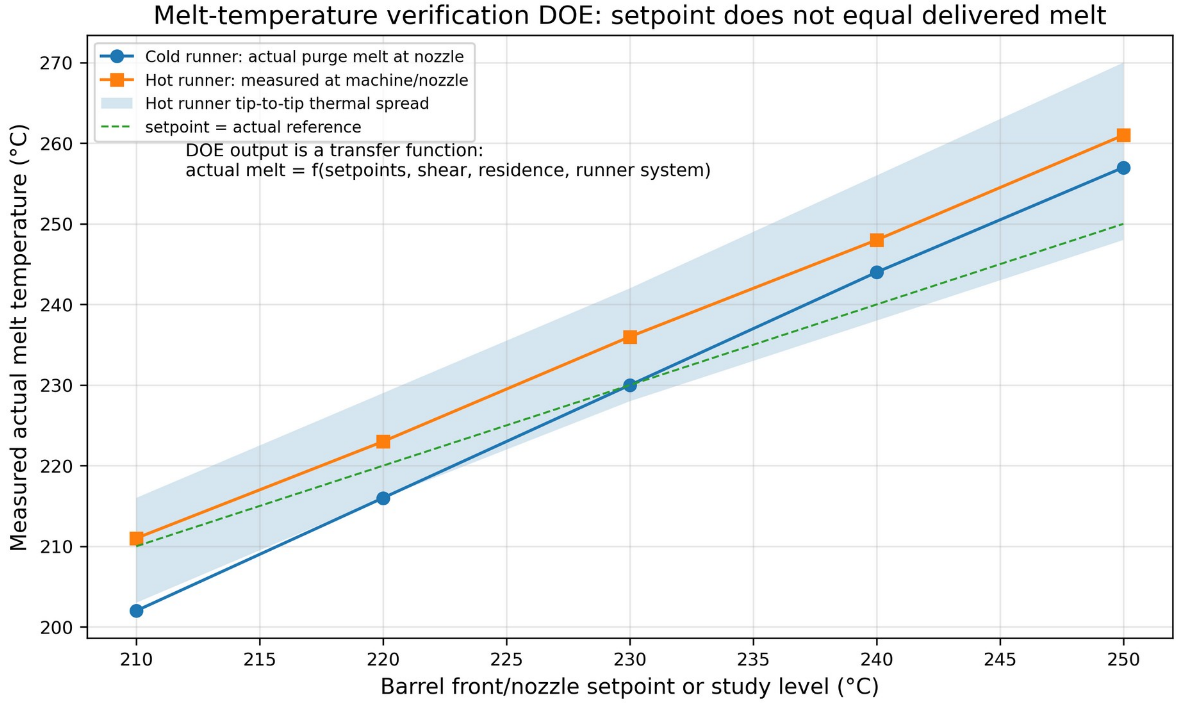

Figure 1. Example output from a melt-temperature verification DOE. Data shown is representative training data, not universal material data.

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Prepared: 2026-05-15

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1. Definition and Overview

A melt temperature verification DOE is a controlled, documented experiment that quantifies the actual polymer melt temperature delivered by an injection molding system under defined production conditions. It maps machine setpoints and process variables to measured melt temperature and product-response variables, then converts that knowledge into a robust process window, setup standard, and reaction plan. The word "verification" is critical: the study is not a barrel-setpoint audit. It verifies the physical temperature of the plastic that actually leaves the nozzle, enters a cold runner, or reaches a hot runner manifold/tip/gate system.

The study is normally run when a mold is being qualified, transferred, revalidated, repaired, converted from cold runner to hot runner, challenged by lot-to-lot resin changes, or suffering defects that point toward viscosity, degradation, melt imbalance, or uncontrolled thermal history. The response variables usually include actual melt temperature, fill time, transfer pressure, peak injection pressure, cavity pressure integral, part weight, dimensions, short-shot pattern, visual defects, weld-line severity, and any critical-to-quality measurement tied to the part print or control plan.

Critical distinction
A melt-temperature verification DOE is not the same as a viscosity curve. The melt-temperature DOE asks: "What actual melt temperature is being delivered, and what machine, material, mold, runner, and auxiliary variables drive it?" A viscosity/rheology curve asks: "At a known thermal state, what fill rate produces the least sensitivity to viscosity variation without creating shear defects?" They are connected, but combining them casually can produce false conclusions.

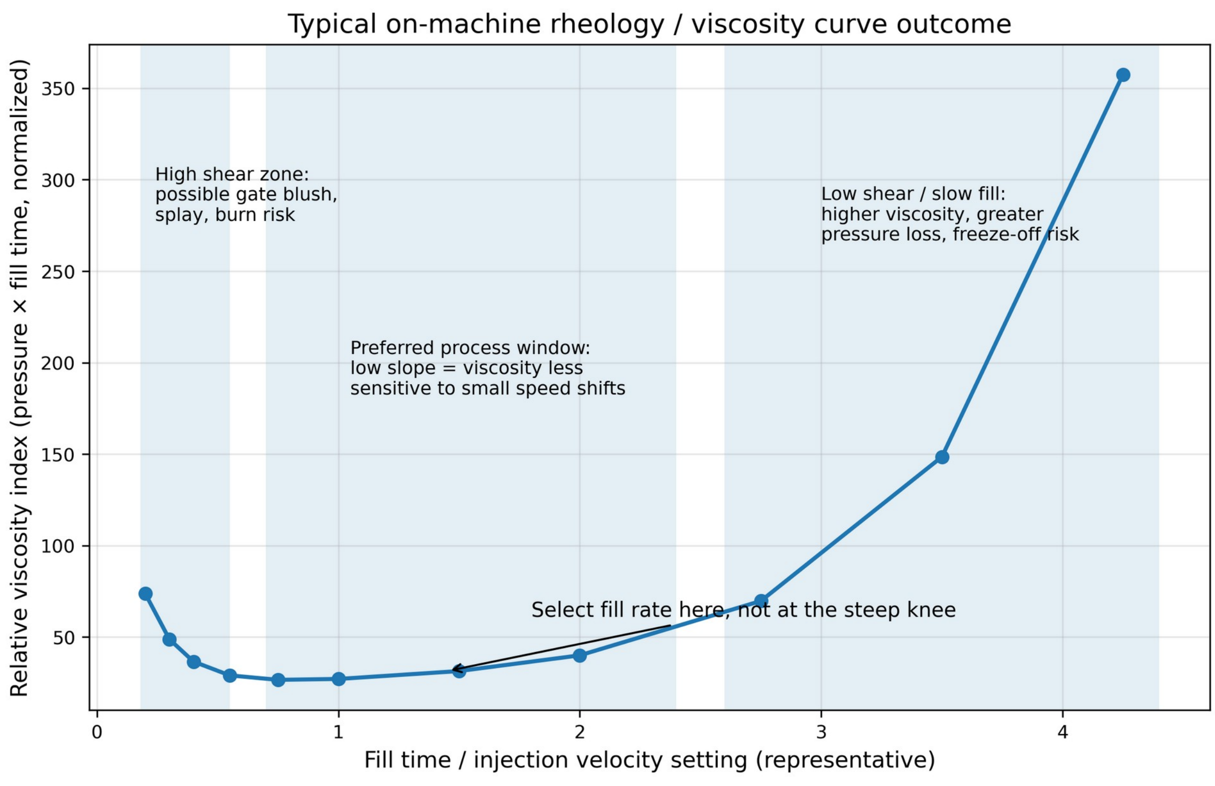

Figure 2. Typical viscosity/rheology curve shape. It teaches fill-rate sensitivity, not actual melt temperature by itself.

1.1 Operational definition

Operationally, the DOE should define the resin grade and lot, dryer condition, machine and screw, shot size ratio, residence time, barrel profile, screw rpm, back pressure, decompression, injection velocity profile, V/P transfer, hold profile, mold temperature, water-flow condition, runner system condition, and sampling method. The result is a transfer function between controllable inputs and actual melt temperature. For a cold runner, that function is dominated by barrel, nozzle, screw work, residence, cold sprue/runner heat loss, and gate pressure loss. For a hot runner, the transfer function also includes manifold zoning, drop/tip control, thermocouple placement, valve-pin conduction, gate design, drool/stringing behavior, and cavity-to-cavity thermal balance.

The most common production error is assuming that a controller setpoint equals the real melt temperature. Research and field guidance repeatedly warn against that assumption. Melt temperature can deviate from band-heater settings and can fluctuate within the cycle due to plasticizing energy, screw speed, back pressure, residence, material heat capacity, shear heating, compression heating, and heat transfer to steel. Modern research has shown that melt temperature measurement is difficult, that conventional sensors can be too slow, and that machine-learning approaches have been proposed because direct sensing is not always production-practical [S9].

1.2 Primary deliverables

Deliverable	What it must contain	Why it matters
Verified melt-temperature range	Actual measured melt temperature across approved machine/mold/resin conditions.	Prevents setpoint-based process transfer errors.
Measurement method	Probe/sensor type, calibration status, purge mass, timing, insertion depth, number of replicates, safety method.	Without measurement discipline the DOE is noise, not data.
DOE matrix	Factors, levels, randomized or blocked sequence, constants, sample size, data sheet, acceptance criteria.	Allows root-cause separation instead of tribal troubleshooting.
Cold/hot runner qualification notes	Cold runner pressure loss and freeze behavior; hot runner manifold/tip zone map and cavity balance.	Runner system changes the thermal and rheological story.
Process optimization decision	Selected barrel profile, screw speed, back pressure, fill rate, hot runner zones, mold temperature, and alarms.	Converts the study into repeatable production control.
Reaction plan	What to do when actual melt temperature, pressure, fill time, part weight, or cavity balance drifts.	Prevents the same problem from recurring.

2. Root Causes: Why a Melt Temperature Verification DOE Is Done

The DOE is performed because melt temperature is a high-leverage process variable that is often indirectly controlled. It influences viscosity, pressure loss, shear heating, molecular orientation, weld-line strength, crystallinity, shrinkage, warpage, gloss, degradation, color stability, and dimensional repeatability. The DOE is also performed because the same barrel setpoints can produce different actual melt temperatures on different machines, with different screws, with different shot-size ratios, or with hot runner systems that add their own thermal dynamics.

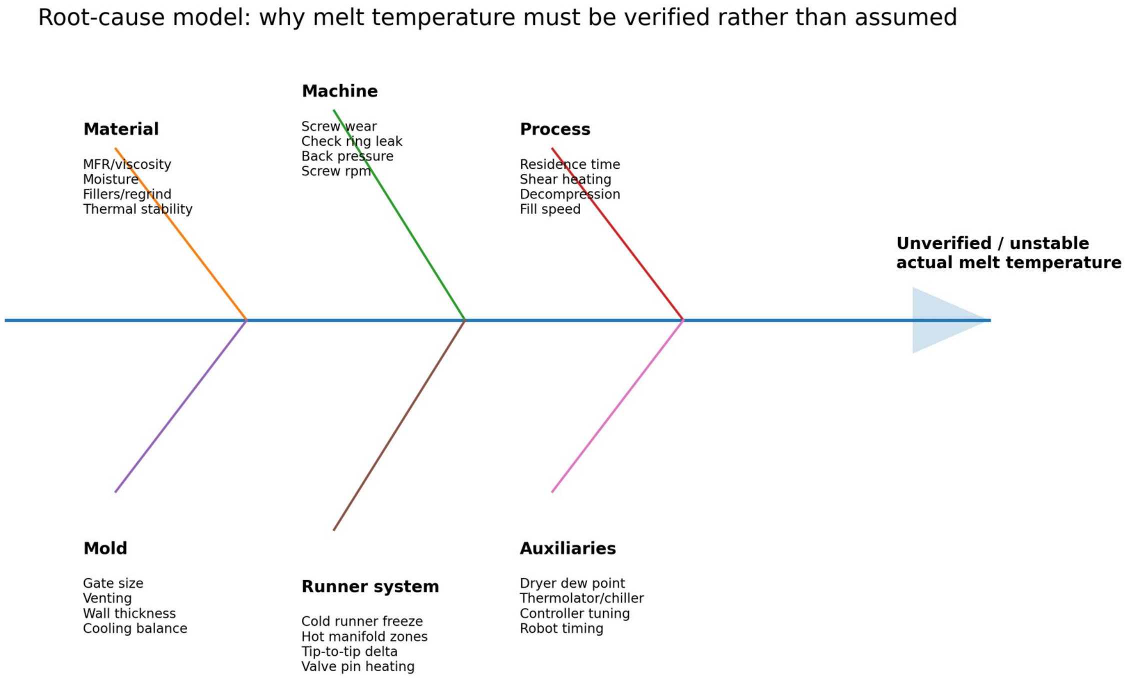

Figure 3. Root-cause model for unverified or unstable actual melt temperature.

2.1 Material factors

Material is not a passive input. Polymer viscosity is a function of shear rate, temperature, pressure, molecular weight distribution, moisture, filler loading, regrind level, pigments, flame retardants, lubricants, nucleators, impact modifiers, and thermal history. ASTM D3835 is specifically intended to measure rheological properties at various temperatures and shear rates common to processing equipment, and ASTM notes that the method is sensitive to molecular weight, stability, additives, plasticizers, lubricants, moisture, reinforcements, and fillers [S2]. A one-point melt flow rate, such as ASTM D1238, is useful for incoming quality control but is not a complete processing rheology model [S10].

Material variable	Effect on melt-temperature DOE	Cold runner impact	Hot runner impact
MFR / MFI / molecular weight	Higher MFR usually lowers apparent viscosity at standard test condition; actual molding response still depends on shear and temperature.	Lower viscosity may hide undersized runner/gate issues until colder conditions or regrind are introduced.	Lower viscosity can worsen drool, stringing, valve-gate leakage, and gate vestige issues if tips are hot.
Moisture	Can cause hydrolysis in susceptible polymers and apparent viscosity loss; can also produce splay and bubbles mistaken for high melt temperature.	Cold runner may show splay in sprue/runner and part; runners can trap evidence for inspection.	Hot runners can retain degraded/moist material longer; purging the manifold is harder than purging a barrel.
Fillers and glass fiber	Increase shear sensitivity, thermal conductivity changes, screw wear, and pressure demand; fiber orientation may be affected by melt temperature and fill rate.	Higher runner/gate shear can break fibers and raise pressure loss.	Tip and gate shear can become the controlling degradation point; fiber hang-up can amplify cavity imbalance.
Lubricants / flow promoters	Can reduce pressure and mimic a warmer melt; can affect screw recovery and cushion consistency.	May reduce cold-runner pressure loss but can increase flash sensitivity.	May change valve-gate seal behavior and drool tendency.
Pigments / masterbatch	Can alter thermal absorption, dispersion, and degradation	Color streaks may show cold slug or poor mixing in	Color changes can expose dead spots in manifolds and tips.

	behavior; dark colors may heat differently under shear and during purge measurement.	sprue/runner.	
Regrind	Variable molecular weight and contamination can distort DOE response; regrind may have higher MFR if degraded.	Runner regrind from cold runner must be controlled or excluded during validation.	No cold runner regrind exists, but startup scrap and manifold hold-up still matter.

Polymer family	Thermal/rheology behavior relevant to DOE	Expected study behavior	Special cautions
PP	Semi-crystalline, shear-thinning, broad commercial MFR range. Processing guides recommend actual melt checks for setup [S13].	Often forgiving, but actual melt and fill rate strongly affect shrinkage, gloss, weld lines, and cycle balance.	Do not use high MFR as a substitute for proper gate/runner sizing. Nucleated or filled PP can shift cooling and shrink response.
ABS	Amorphous, generally good processing window but sensitive to shear/thermal degradation, moisture-related splay, and weld-line cosmetics.	Viscosity curve often identifies a stable fill-rate plateau; melt-temperature DOE exposes color/gloss shifts and burn sensitivity.	Excessive shear or high residence can yellow or degrade; venting must be verified before raising temperature or speed.
PC	Amorphous, high melt temperature, high viscosity, moisture-sensitive, high pressure demand.	Small errors in actual melt temperature can create large pressure and stress changes; drying and residence control are critical.	High hot runner residence or dead spots can cause splay, black specks, silvering, and property loss.
PA / Nylon	Semi-crystalline, moisture-sensitive, strong mold-temperature effect on crystallinity and dimensions.	Drying and mold thermal balance are central. Higher mold temperature can improve surface/weld performance but changes shrink.	Hydrolysis can masquerade as better flow; verify moisture before attributing lower pressure to higher melt temperature.
POM / Acetal	Sharp melting behavior and degradation safety risk if overheated or contaminated.	DOE must stay inside resin supplier limits and residence limits; actual melt temperature is a safety-critical variable.	Use purging and temperature controls specified by resin supplier; avoid incompatible contamination.
Glass-filled engineering resins	Higher pressure demand, shear heating, abrasive screw wear, and anisotropic shrinkage.	DOE should include screw recovery energy and screw/barrel condition; pressure response may drift as hardware wears.	Hot tips and gates can become fiber filters; cavity balance must be verified with short shots and part weights.

2.2 Process parameters

Process parameters change both the generated heat and the rate at which heat is lost. Injection speed changes shear rate. Back pressure and screw rpm change plasticizing work. Residence time changes heat soak and degradation risk. Mold temperature changes how fast the melt freezes, which changes pressure loss, skin thickness, weld-line formation, gloss, and the interpretation of short shots. Fill time affects shear heating, shear thinning, pressure, temperature inside cavities, and part quality, and Beaumont/Plastics Technology emphasize documenting plastic variables such as actual melt temperature instead of only machine setpoints [S3].

Parameter	DOE effect mechanism	Typical response if increased	Risk if misinterpreted
Barrel/nozzle setpoints	External heater input plus thermal profile around screw.	Actual melt may rise, but not one-to-one; response can lag and differ by zone.	Changing front zone only may not correct poor melting or excessive shear heat generated upstream.
Screw rpm	Mechanical shear work during recovery.	Actual melt temperature often rises; recovery time falls.	A pressure drop after higher rpm may be due to shear heating, not better material.

Back pressure	Adds work and mixing during plasticizing.	More homogeneous melt and higher actual melt temperature, but longer recovery and more shear.	Too much back pressure can degrade shear-sensitive materials or break fibers.
Injection speed / fill rate	Changes shear rate during filling and viscous heat generation.	Pressure may fall in shear-thinning region even though velocity is higher.	A lower pressure does not automatically mean a safer process; shear burns and gate blush may increase.
Cooling time	Changes mold thermal equilibrium and part ejection temperature.	Longer cycles may cool hot runner gates or raise/lower steel equilibrium depending on system.	DOE run before thermal stabilization produces non-reproducible results.
Mold temperature	Controls heat extraction, skin formation, flow length, crystallinity, gloss.	Higher mold temp can lower pressure and improve surface/weld appearance.	Mold-temperature changes can be mistaken for melt-temperature effects.
Transfer position	Changes fill-only shot volume and pressure data.	Later transfer may pack during a viscosity study and corrupt the curve.	A viscosity curve must normally be fill-only/short-shot, not packed.

2.3 Mold design factors

A mold can make a good melt look bad. Runner length, gate thickness, venting, wall thickness, cooling layout, steel mass, cavity count, and flow length all determine how the melt loses heat and pressure after leaving the barrel. Moldex3D and Autodesk Moldflow workflows rely on shear viscosity, PVT, thermal conductivity, heat capacity, and process settings because these variables directly affect sprue pressure, melt-front advancement, cooling, shrinkage, and warpage predictions [S6, S7].

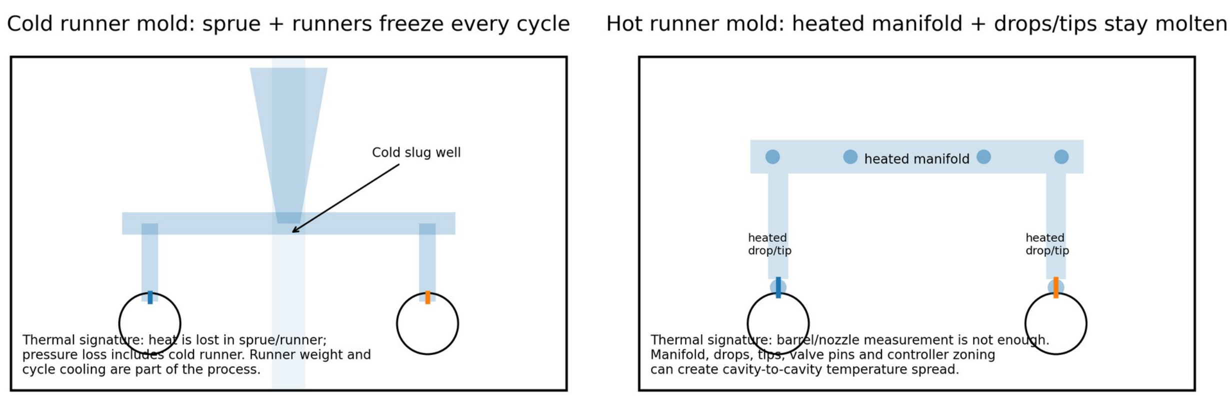

Figure 4. Cold runner and hot runner thermal architecture. The verification plan must be different for each system.

Mold issue	Effect during DOE	Cold runner signature	Hot runner signature
Undersized gate	Raises shear rate, pressure loss, and local viscous heating.	Gate blush, high pressure at short fill, freeze-off, high runner-to-part pressure ratio.	Tip/gate overheats or degrades while manifold appears correct.
Gate too large	Can reduce shear and pressure but increase vestige, drool, gate blush, long gate seal.	Larger cold runner/gate may extend cycle and increase regrind.	Valve-gate timing and seal become critical; drool/stringing may rise.
Poor venting / air traps	Burns and shorts may be blamed on melt temperature when gas evacuation is the true limit.	Burns at end-of-fill; pressure spike late in fill.	Cavity-specific burns may align with individual hot drops or gate sequencing.
Wall-thickness variation	Thick-to-thin transitions change flow speed, freeze rate, and pressure demand.	Short-shot pattern may show hesitation and race-tracking through thick sections.	Sequential valve gating can hide or amplify race-tracking.
Cooling imbalance	Changes steel temperature during DOE and creates	Core/cavity side differences show as warp and gloss	Gate area cooling may fight the hot tip and create stringing,

	misleading temperature/pressure trends.	variation.	cold slugs, or freeze-off.
Cavity imbalance	Multi-cavity data may average away worst-cavity behavior.	Runner layout, shear-induced imbalance, and gate land variation dominate.	Manifold balance, tip thermocouple location, and valve-pin timing dominate.

2.4 Machine factors

Machine condition changes the relationship between setpoint and actual melt. Screw design, L/D ratio, compression ratio, mixing section, non-return valve leakage, barrel wear, nozzle type, heater-band condition, thermocouple contact, hydraulic/electric injection response, clamp stiffness, and controller tuning all affect the experiment. A process transferred from one machine to another can fail even when the recipe is copied if the plastic variables are not matched.

Machine factor	Why it matters	Study control / diagnostic
Screw design	General-purpose, barrier, mixing, low-compression, and high-compression screws generate different melting and shear profiles.	Record screw type, diameter, L/D, compression ratio, mixing section, and material compatibility.
Screw/barrel wear	Wear increases leakage, unstable recovery, poor melting, and shot-to-shot temperature variation.	Compare recovery time, cushion stability, check-ring leak test, and melt homogeneity. Inspect on maintenance interval.
Back pressure control	Actual hydraulic/electric pressure may not equal controller setting; drift alters mechanical heat.	Verify actual back pressure and recovery time during every DOE run.
Nozzle design	Open nozzle, shut-off nozzle, mixing nozzle, extended nozzle, or hot sprue interface changes residence and heat loss.	Document nozzle style, tip radius, orifice, heater status, and contact with sprue bushing/hot sprue.
Clamp force	Clamp force does not directly set melt temperature but affects flash, vent closure, parting-line gas escape, and perceived viscosity limits.	Do not solve flash caused by over-pack/high melt with excessive clamp; verify clamp tonnage, mold protection, and vent condition.
Injection response	Velocity linearity and pressure-limit margin determine whether the commanded fill-rate points are real.	Confirm not pressure limited during viscosity curve. Keep at least a practical pressure reserve.

2.5 Auxiliary equipment factors

Auxiliary equipment can create false DOE results if treated as background noise. The dryer sets moisture and material temperature. Chillers and thermolators set mold thermal equilibrium. Hot runner controllers set manifold and tip temperatures. Robots and sprue pickers affect cycle repeatability, open-mold time, runner handling, part cooling, and whether short shots are sampled consistently.

Auxiliary	Role in melt-temperature DOE	Cold runner emphasis	Hot runner emphasis
Dryer / hopper loader	Moisture, material preheat, and contamination control.	Runner regrind addition must be controlled; dryer drawdown and open bins matter.	Startup and hot-runner residence magnify moisture/degradation issues.
Chiller / thermolator	Controls mold surface temperature and thermal stabilization.	Runner cooling changes pressure loss and cycle time.	Gate cooling must balance tip heat; water-flow restriction can mimic hot-runner temperature error.
Hot runner controller	Controls zones, alarms, thermocouple inputs, heater outputs.	Not applicable except heated sprue bushing or hot-to-cold hybrid.	Central to DOE. Log actual zone PV/SV, output %, alarms, zones grouped, thermocouple type, and tuning state.

Robot / sprue picker	Affects cycle repeatability and sampling timing.	Sprue picker delays can cool runners before weighing; runner separation timing matters.	Robot delay affects mold-open time and gate area cooling; sequential sampling by cavity must be disciplined.
Gravimetric blender / regrind feeder	Changes material composition and apparent viscosity.	Cold runner regrind percentage can drift with grinder performance.	No runner regrind, but startup scrap and colorant dosing remain important.

3. DOE Architecture and Measurement System

A melt-temperature verification DOE must be built like a measurement-system and process-window study. The usual sequence is: stabilize the cell, verify measurement method, run a baseline, vary selected factors, collect actual temperature and quality responses, model main effects and interactions, select a robust window, and validate it under production conditions. The study should avoid changing multiple uncontrolled variables between data points. If a maintenance correction is made during the DOE, block the data or restart the matrix.

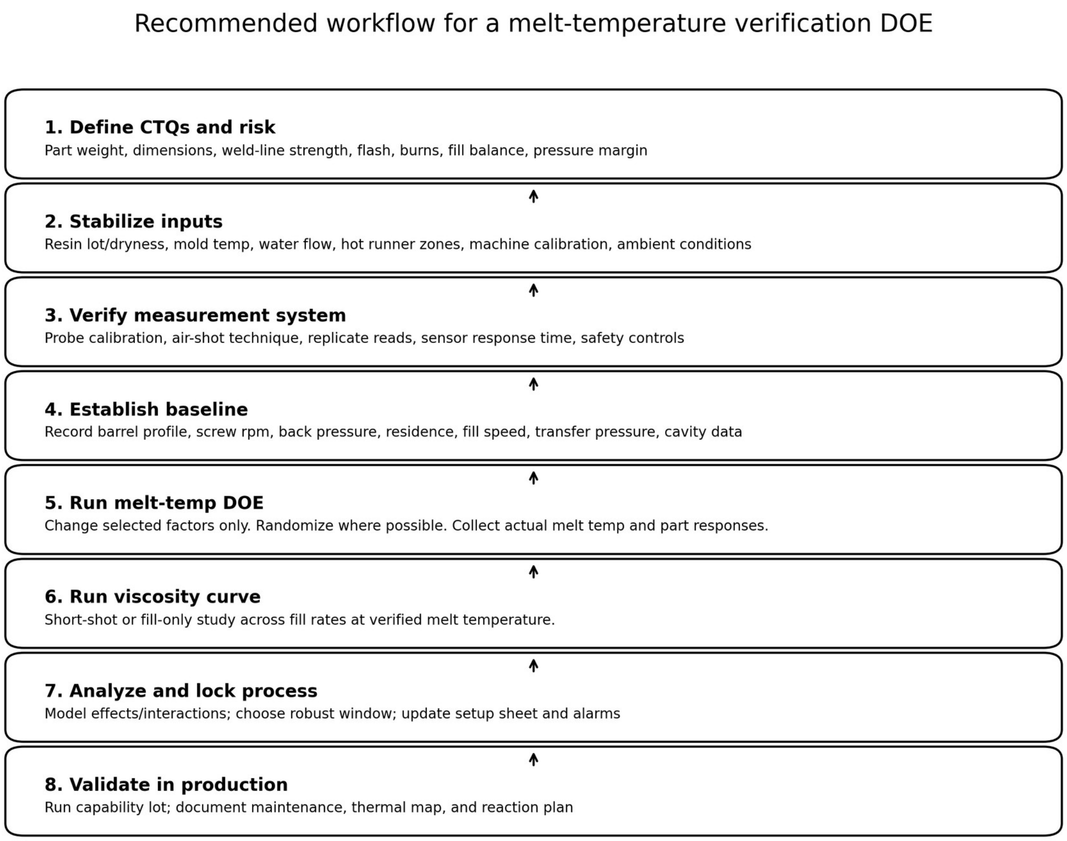

Figure 5. Workflow for running a controlled melt-temperature verification DOE.

3.1 Measurement methods

Measurement method selection depends on safety, access, resin temperature, resin degradation risk, and whether the mold is cold runner or hot runner. Air-shot measurement with a fast needle probe is common and low cost, but it is technique-sensitive. Inline nozzle sensors, infrared sensors, in-mold temperature sensors, and specialized melt-temperature systems can improve response time or repeatability. Research notes that sensors must survive high temperature and pressure and that conventional sheath thermocouples may be too slow for dynamic melt-temperature measurement [S9].

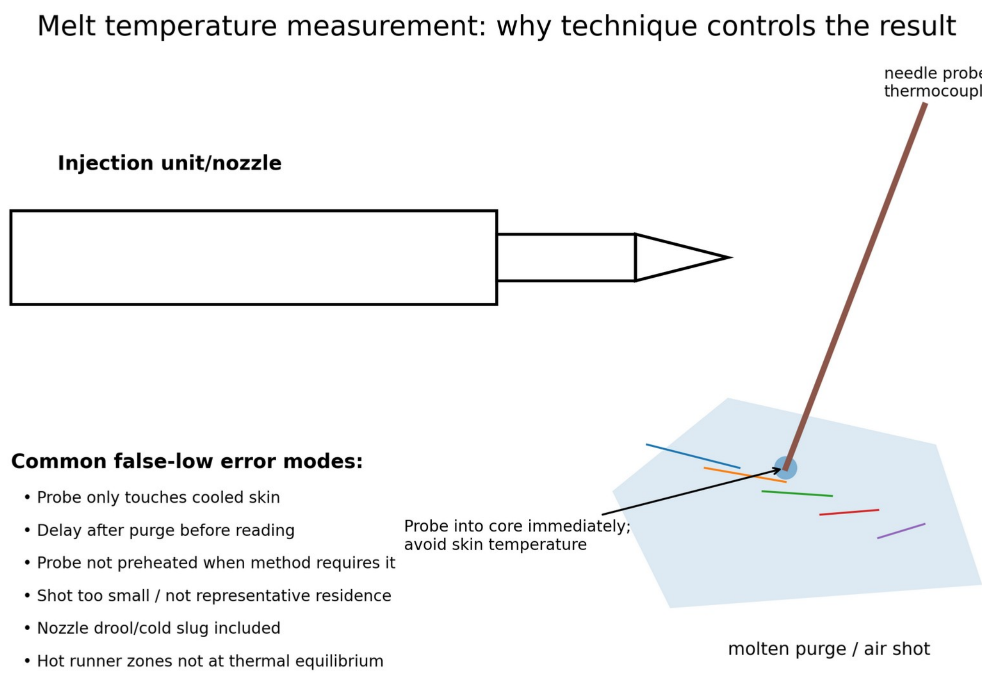

Figure 6. Air-shot measurement and common false-low error modes. Use EHS-approved guarding and PPE.

Method	Best use	Advantages	Limitations / false conclusions
Air shot into insulated catch area + needle probe	Setup, qualification, troubleshooting, low-cavitation tools.	Low cost, fast, works on most machines.	Reads cooled skin if delayed; operator technique affects data; not a dynamic cycle measurement.
Nozzle adapter thermocouple or infrared sensor	Validation, research, high-value tools, process monitoring.	Can capture dynamic temperature profile nearer production cycle.	Adds hardware, may affect flow, may require machine/nozzle modification and calibration.
Hot runner manifold/tip thermocouples	Hot runner qualification and zone diagnostics.	Available every cycle from controller; good for zone alarms.	Thermocouple position measures steel, not always melt. Equal zone readings do not prove equal melt delivery.
In-cavity temperature sensors	Scientific molding, medical validation, multi-cavity balance.	Captures thermal arrival and cavity behavior.	Measures at sensor location after heat transfer to steel/melt interface; not the same as bulk melt temperature.
Capillary rheometer	Material characterization, supplier data validation, simulation inputs.	Measures viscosity across temperature and shear rate; ASTM D3835 is built for processing-relevant shear rates [S2].	Not the same as the full machine/mold system; does not include screw, runner, gate, or mold effects.
MFR / MVR testing	Incoming material QC, degradation screening.	Simple quality-control measure under ASTM D1238 [S10].	One-point empirical result; should not be treated as the full viscosity curve.

3.2 Measurement-system verification

1. Calibrate or verify the temperature probe against a traceable reference at the expected process range. Ice-water checks are not enough for a 250-320 °C melt process.

2. Define purge volume, purge duration, and measurement delay. Record time from purge to first reading.

3. Measure the core of the purge, not the cooled surface skin. For high-temperature resins, an insulated receiving method is safer and more repeatable than an open pile on a cold table.

4. Use at least three replicate readings at each DOE condition. If replicate spread exceeds your engineering tolerance, solve measurement variation before running the DOE.

5. Record machine cycle state. A measurement after a long idle, alarm, nozzle contact break, or hot-runner soak is not equivalent to steady production.

6. For hot runners, log barrel/nozzle actuals, manifold zone PV/SV, tip zone PV/SV, controller output %, water temperatures, and cavity-specific results. Do not use a single barrel air-shot value as the only hot-runner evidence.

3.3 DOE factor selection

DOE type	When to use it	Factors	Notes
Screening DOE	Unknown process, high scrap, many suspected drivers.	Barrel profile, screw rpm, back pressure, residence, mold temp, hot runner tip delta.	Use two levels plus center points. Keep material lot and dryer state fixed.
Response-surface DOE	Need optimized setpoint, not just direction.	Melt setpoint, screw rpm, back pressure, mold temp.	Useful when actual melt target must be achieved without degrading resin.
OFAT verification	Low-risk production verification or quick confirmation after maintenance.	One factor at a time.	Simple, but weak for interactions. Acceptable only when the interaction risk is low.
Nested cavity / hot-runner DOE	Multi-cavity or multi-drop hot runner imbalance.	Drop/tip zone, cavity, valve timing, manifold branch.	Cavity is a factor. Do not average away worst cavity behavior.
Transfer DOE	Process moved to a different press or duplicate mold.	Machine, screw, nozzle, thermal profile, fill rate.	Use plastic variables as matching targets, not only machine settings.

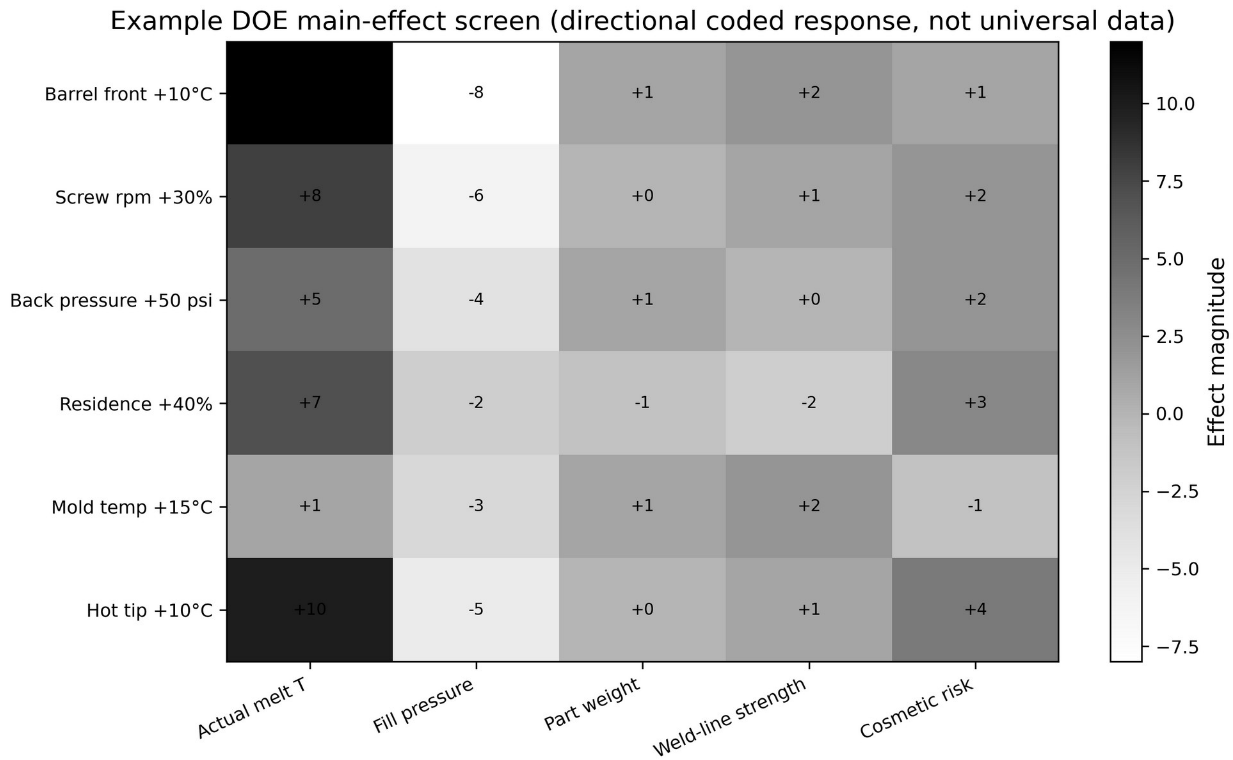

Figure 7. Example coded main-effect screen. Direction and magnitude are illustrative; real results must come from the mold/resin/machine.

3.4 Minimum data sheet

Field	Required data
Identification	Part number, mold number, revision, cavity count, resin grade/lot, colorant lot, regrind %, operator, processor, date/time.
Machine	Press ID, screw diameter/design, nozzle, shot size, cushion, recovery time, check-ring status, clamp force, pressure limit.
Material prep	Dryer model, dew point, drying temperature/time, hopper residence, moisture test if applicable.
Thermal inputs	Barrel zone SV/PV, nozzle SV/PV, mold water supply/return, flow rate, hot runner manifold/drop/tip PV/SV/output %.
Process inputs	Screw rpm, back pressure, decompression, fill speed profile, transfer position/pressure, hold pressure/time, cooling time, cycle time.
Responses	Actual melt temp replicates, fill time, transfer pressure, peak injection pressure, cavity pressure, part weight by cavity, dimensions, visual grade, defect notes.
Environment	Ambient temperature/humidity, material exposure, start-up/steady-state status, maintenance changes, alarms.

4. Viscosity-Curve Outcomes and Interpretation

The viscosity curve, also called an on-machine rheology curve, is normally produced by running fill-only or short-shot conditions at several fill rates and plotting a relative viscosity index versus fill time, injection velocity, or relative shear rate. RJG describes effective viscosity in practical molding terms as the work required to flow material, commonly approximated from fill time multiplied by pressure at transfer [S1]. Beaumont/Plastics Technology describe fill time as affecting shear heating, shear thinning, viscosity, pressure, temperature inside cavities, dimensions, aesthetics, and strength [S3].

A good curve usually has a steep region at slow/low-shear fill, a lower-slope plateau where small speed variation causes less viscosity variation, and sometimes a high-shear region where gate blush, burning, fiber breakage, material degradation, or excessive pressure rise becomes unacceptable. The selected fill rate should normally come from a stable region that meets quality requirements and keeps the machine out of pressure limitation. It should not be chosen only because it produces the lowest pressure. A very fast fill can reduce apparent viscosity through shear thinning while simultaneously damaging the gate, burning trapped air, or over-heating the melt locally.

Curve observation	Likely interpretation	Action
No clear plateau; pressure rises continuously as speed increases	Material/tool may be shear sensitive, gate undersized, or machine pressure limited.	Check pressure limit, gate shear, runner/gate sizing, melt temp, venting, and machine velocity response.
Very steep curve at slow fill	High viscosity and excessive heat loss during fill.	Avoid slow fill unless cosmetic/venting constraints demand it; verify pressure reserve.
Pressure falls sharply as speed increases	Shear thinning dominates; warmer localized melt may flow easier.	Select stable low-slope range; inspect for blush, burns, splay, jetting, and fiber break.
Curve shifts upward after resin lot change	Higher viscosity, lower actual melt, moisture/drying change, or more filler/regrind.	Verify actual melt temperature and material prep before adjusting pack/hold.
Hot runner cavities curve differently	Tip/manifold imbalance, valve timing, cavity fill differences, or thermocouple/control issue.	Analyze by cavity/drop. Do not rely on averaged transfer pressure alone.
Cold runner curve changes as cycle warms up	Sprue/runner and mold thermal balance not stabilized.	Discard startup data; run DOE after thermal equilibrium.

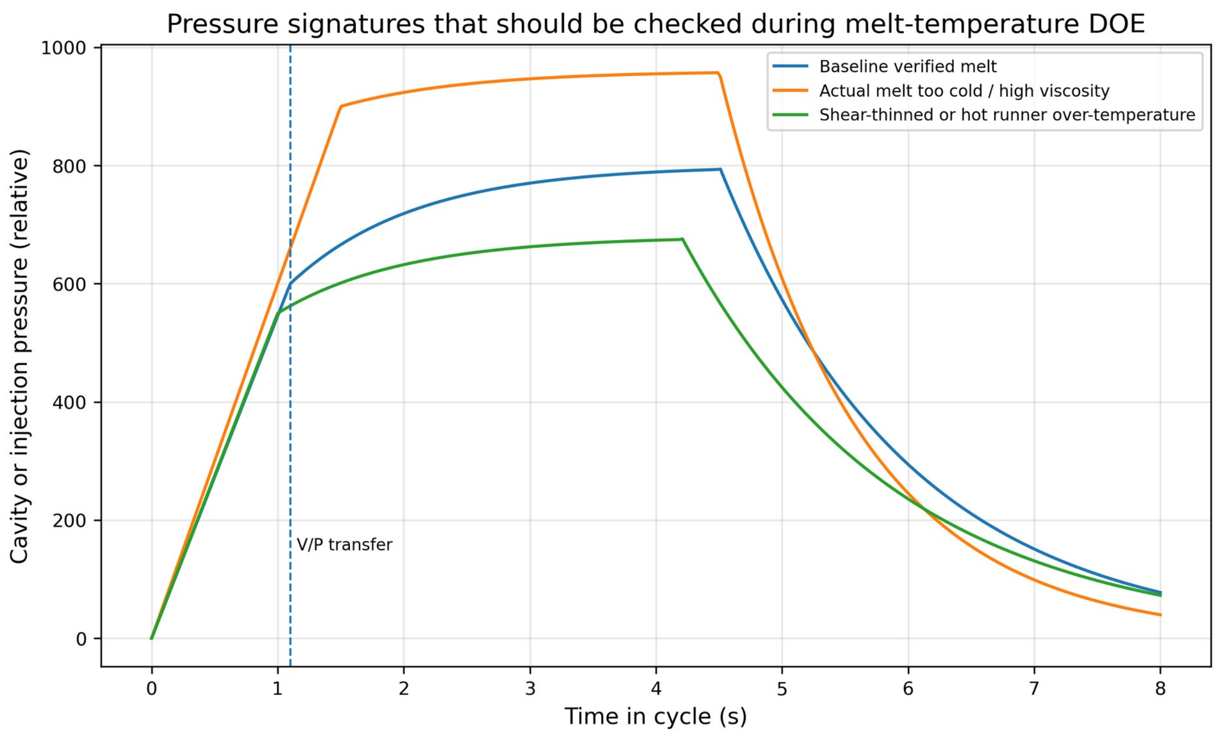

Figure 8. Pressure signatures that should be reviewed during melt-temperature and viscosity studies.

4.1 Relative viscosity formula and cautions

A practical shop-floor metric is: relative viscosity index = fill time x pressure at transfer. More advanced systems use the area under the injection pressure curve during fill or cavity pressure integrals. The metric is relative; it is not the same as laboratory viscosity in Pa·s. It should be used to compare controlled molding conditions on the same machine/mold/material combination, not as a universal material property.

False-data warning
Do not include pack/hold in the fill-only viscosity data. Do not allow the machine to pressure-limit at high speed. Do not compare data taken at different actual melt temperatures unless the study deliberately includes temperature as a factor. Do not average cavities in a multi-cavity hot runner until the worst-cavity behavior has been reviewed.

5. Hot Runner Versus Cold Runner Rules

The difference between hot runner and cold runner molds is not a footnote; it changes the DOE architecture. A cold runner mold freezes a sprue/runner/gate system every shot. The measured melt leaving the nozzle is reasonably close to the melt entering the sprue, but the plastic loses heat and pressure through the cold runner before the cavity. A hot runner mold keeps a manifold/drop/tip system molten. The barrel/nozzle measurement alone does not prove what each cavity receives because the manifold and tips add heat, residence, thermal gradients, and cavity-specific imbalance. Husky emphasizes hot-runner balanced flow, gate quality, and multi-drop configurations; hot-runner balance is therefore part of mold qualification, not an optional add-on [S18, S19].

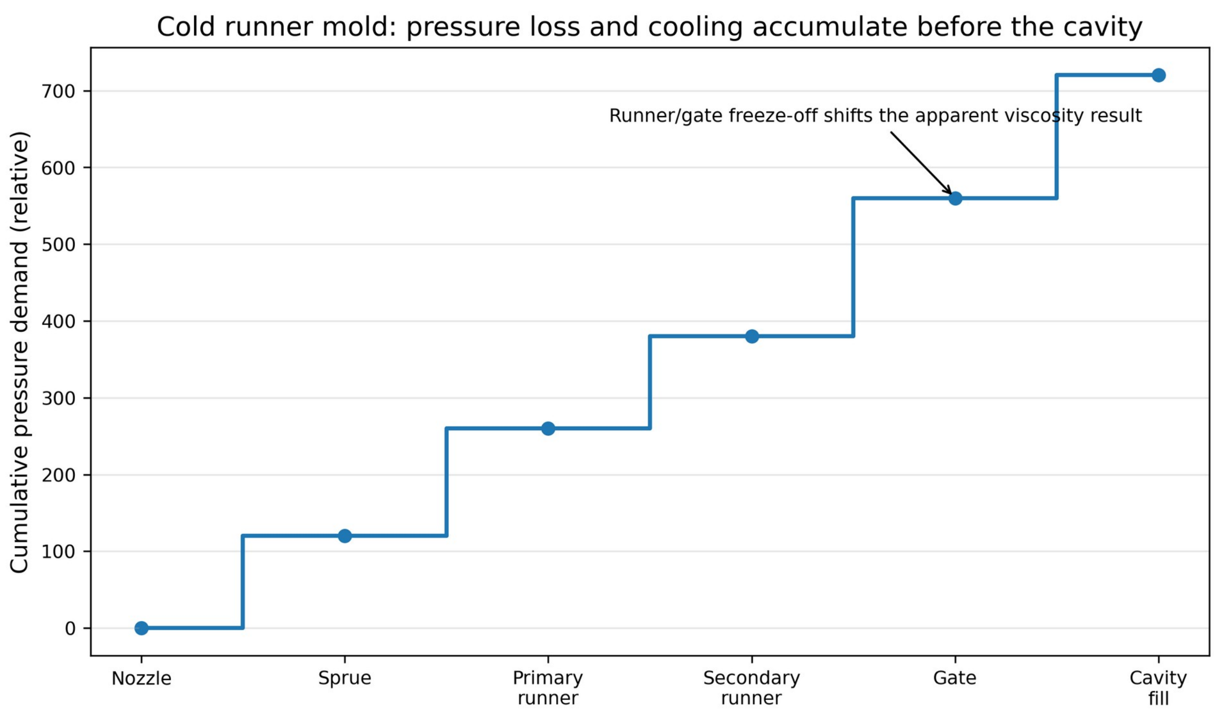

Figure 9. Cold runner pressure loss accumulates through the feed system and can bias viscosity interpretation.

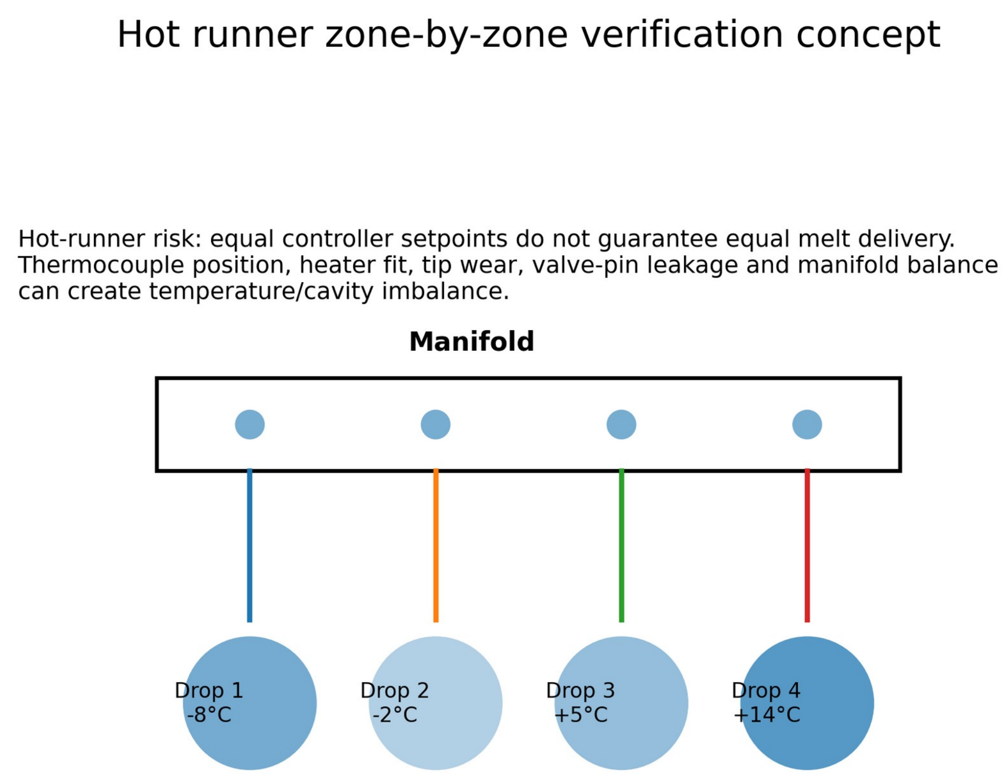

Figure 10. Hot runner verification must include zone-by-zone and cavity-by-cavity evidence.

Topic	Cold runner DOE requirement	Hot runner DOE requirement
Melt-temperature measurement	Air-shot/nozzle actual temperature is highly relevant, but runner heat loss must be considered.	Measure machine/nozzle, log manifold/tip zones, verify cavity balance. A nozzle reading alone is insufficient.
Stabilization	Run until mold, sprue puller, runner cooling, and part weight stabilize.	Run until manifold, tips, valve pins, gate area cooling, and controller outputs stabilize.
Short-shot study	Include runner fill and part fill. Weigh runners and parts separately if needed.	Short-shot by cavity/drop; record valve timing and which zones feed each cavity.
Regrind	Runner regrind percentage is part of the material system and must be fixed or excluded.	No runner regrind, but startup scrap and manifold residence must be controlled.
Pressure loss	Sprue/runner/gate geometry can dominate pressure and freeze behavior.	Manifold/drop/tip/gate thermal balance can dominate cavity balance and degradation.
Common false fix	Raising melt temperature to overcome an undersized cold runner or gate.	Raising all hot runner zones to fix one cold tip, overheating other drops.
Maintenance link	Inspect sprue bushing, runners, gates, vents, cooling, parting line.	Inspect heaters, thermocouples, tip fit, seals, valve pins, wiring, controller tuning, insulation, cooling around gates.

5.1 Cold runner study sequence

1. Confirm sprue bushing/nozzle match, clean nozzle tip, correct sprue puller function, and no drool or cold slug obstruction.

2. Stabilize mold temperature and runner ejection timing. Use consistent runner handling and weighing method.

3. Run melt-temperature verification at the nozzle under steady cycle, not after a long purge or alarm unless that is a defined startup test.

4. Run pressure-loss ladder if needed: air shot, sprue-only, runner-only, gate/cavity short shot, and full fill. The goal is to locate the pressure/thermal restriction.

5. Run viscosity curve only after the actual melt temperature and mold temperature are stable.

6. Set fill rate, transfer, pack, cooling, and regrind control from the verified process window.

5.2 Hot runner study sequence

1. Verify hot runner nameplate/specification, resin compatibility, heater/thermocouple wiring, zone mapping, controller tuning, and safe operating temperature window.

2. Bring manifold/tip zones to thermal equilibrium. Record setpoint, actual, output %, and alarms. Grouped zones should be treated as higher risk than independently controlled tips.

3. Measure actual barrel/nozzle melt under steady production. Then evaluate hot runner delivery through short shots, cavity weights, cavity pressure, gate appearance, and thermal balance.

4. For multi-cavity systems, collect cavity-specific data. Do not accept an average part weight if one cavity is cold, hot, delayed, degraded, or flashing.

5. Run tip/manifold delta checks carefully. Changing all tips by the same amount is different from balancing one drop.

6. Validate startup, interruption, and restart behavior because hot runner residence can create degraded material that is not represented by steady-state data.

6. Real-World Examples, Case Patterns, and Practitioner Accounts

Evidence standard for this section
No private interviews were conducted for this document. The practitioner comments below are based on public forum-style observations and are treated as anecdotal, not validated evidence. The case studies are realistic composite scenarios based on common production patterns and source-grounded principles; they are not presented as confidential accounts from named companies.

6.1 Public practitioner observations

An Eng-Tips thread on melt temperature immediately asks how the melt temperature is being measured and notes that appropriate temperature depends on grade, mold, and machine [S24]. That is the correct diagnostic instinct: before changing process settings, verify the measurement method and system context. A Reddit r/InjectionMolding discussion includes a technician reporting needle-probe melt readings that seem 10-20 degrees short and asking about alternative methods [S25]. The value of that anecdote is not the numeric offset; the value is that real shop-floor measurements often disagree because probe delay, purge cooling, skin formation, and technique dominate the result.

6.2 Composite case study A - Medical closure, PP, cold runner

Problem: A multi-cavity cold runner mold producing child-resistant polypropylene closures showed intermittent short shots in two cavities, rising transfer pressure, and occasional gate blush after a resin lot change. The team first raised barrel temperature 10 °C, which reduced shorts but increased flash and dimensional spread. A melt-temperature verification DOE was run with barrel front/nozzle setpoint, screw rpm, and back pressure as factors. Actual melt temperature, fill time, transfer pressure, cavity weights, and short-shot pattern were recorded. Runner and part weights were separated.

Finding: The initial “low melt temperature” conclusion was partly false. The air-shot method had a delay that allowed the purge skin to cool. After correcting the measurement technique, actual melt temperature was inside the resin window. The real driver was pressure loss through a runner branch and a cold gate land. The viscosity curve identified a stable fill-rate range, but the bad cavities remained pressure-hungry. Tool work corrected the gate land and improved venting. The final process used the verified actual melt temperature, a slightly faster fill rate in the plateau region, fixed regrind percentage, and documented runner weight limits. The lesson is that a viscosity curve cannot fix a geometric pressure-loss defect by itself.

6.3 Composite case study B - Automotive PC/ABS trim, hot runner

Problem: A valve-gated hot runner mold produced gloss variation and weld-line visibility on a cosmetic automotive trim component. Barrel setpoints were within the supplier window, and the nozzle air-shot temperature looked acceptable. The team suspected material, but cavity-specific part weights and visual inspection pointed toward one side of the tool. The DOE added hot runner tip delta, manifold temperature, valve sequence timing, and mold water temperature as controlled factors. Response variables included cavity pressure curves, gate appearance, gloss grade, short-shot flow-front progression, and part weight by cavity.

Finding: A grouped tip zone hid one cold drop and one hot drop. The cold drop generated hesitation and visible weld-line severity; the hot drop created gate blush and occasional odor/black specks after interruptions. Balancing tips individually, correcting a thermocouple seating issue, and adding a restart purge protocol resolved the dominant variation. The lesson is that hot runner verification must be drop-specific; a barrel melt check only verifies the plastic before the hot runner, not the delivered melt at each gate.

6.4 Composite case study C - Consumer goods ABS housing, cold-to-hot conversion

Problem: A consumer goods ABS housing was converted from a cold runner family mold to a hot runner to reduce scrap and cycle time. Initial trials copied the old barrel settings and fill time. The new hot runner produced better short-term fill but more splay and color shift after weekend shutdowns. The melt-temperature verification DOE compared cold-runner baseline data to hot-runner steady-state and restart conditions. The team tracked actual nozzle melt, manifold/tip temperatures, residence time, purge mass, MFR before/after molding, color, and black specks.

Finding: The cold runner had purged the system every shot through the runner, while the hot runner increased residence in heated steel. Startup and interruption behavior became the controlling risk. The final process lowered selected hot runner zones, added material-specific maximum interruption time, increased startup purge volume, and restricted weekend hot-soak. The lesson is that hot runner conversion changes material residence and degradation risk even when the cavity geometry is unchanged.

7. Diagnostic Techniques

7.1 How to identify whether the DOE has been done after production is running

A molded part cannot prove that a melt-temperature verification DOE was performed. Visual inspection, microscopy, or destructive testing can identify symptoms consistent with poor melt-temperature control, but they cannot establish that a DOE was done. The only reliable evidence is documentation: DOE matrix, raw data, setup sheet, calibration records, thermal maps, cavity balance data, capability run, sign-off, and revision-controlled reaction plan. If no records exist, assume the DOE has not been performed to a defensible standard.

Evidence type	What it can show	What it cannot prove
Visual inspection	Splay, silver streaks, burn marks, blush, flow lines, weld lines, gloss variation, shorts, flash.	Cannot prove actual melt temperature or that a DOE was run.
Part weight / dimensions	Process drift, cavity imbalance, pack variation, shrinkage changes.	Cannot isolate melt temperature without controlled data.
Microscopy / cross-section	Voids, weld-line morphology, unmelt, fiber orientation trends, contamination.	Usually cannot distinguish melt-temperature cause from gate/vent/fill-rate cause alone.
DSC	Melting/crystallization behavior, crystallinity trends, material identification support under ASTM D3418-type methods [S11].	Does not reconstruct exact molding melt temperature.
MFR/MVR after molding	Possible degradation or molecular-weight shift if compared to virgin baseline [S10].	Does not identify where degradation occurred: barrel, nozzle, hot runner, or regrind loop.
Cavity pressure history	Fill, pack, gate seal, viscosity shifts, cavity balance. Kistler systems use cavity pressure and mold temperature for monitoring and documentation [S22].	Needs baseline and sensor calibration; not a direct melt-temperature measurement.

7.2 Mold-flow simulation and alternatives

Moldflow, Moldex3D, and similar CAE tools can predict filling pattern, pressure demand, shear rate, temperature distribution, weld lines, air traps, cooling, shrinkage, and warpage. Their accuracy depends on geometry, mesh quality, boundary conditions, material characterization, runner/hot-runner modeling, and validation against short shots and production pressure data. Moldex3D notes that shear viscosity is crucial for calculating sprue pressure, melt-front advancement, and clamping force, and that PVT and thermal properties are needed for shrinkage, packing, warp, and heat-transfer calculations [S6].

Simulation is not a replacement for a melt-temperature verification DOE. It is a pre-production risk-reduction tool and a hypothesis generator. It can identify likely gate restrictions, high-shear zones, weld-line locations, air traps, and thermal imbalance before cutting steel. The DOE confirms what the real machine, screw, runner system, material lot, and plant utilities actually deliver.

Diagnostic method	Use case	Strength	Caution
Moldflow / Moldex3D fill-pack-cool-warp	Predict pressure, fill pattern, welds, air traps, shrink/warp.	Excellent for design-stage decisions and DOE planning.	Needs correct material data, actual process inputs, and validation.
Short-shot progression	Validate flow pattern, balance, hesitation, jetting.	Simple and powerful; reveals real-world flow front.	Must be run at steady thermal state and by cavity/drop.
Pressure-loss study	Locate restriction between nozzle, sprue, runner, gate, and cavity.	Separates machine/nozzle issue from mold feed issue.	Requires disciplined partial-fill method.
Cavity pressure / temperature sensing	Monitor production quality and validate	High-value for medical, automotive, multi-cavity hot	Sensor placement and calibration matter.

	transfer/pack/gate seal.	runners.	
Laboratory capillary rheometry	Material flow characterization across shear and temperature.	More rigorous than MFR for processing-relevant shear [S2].	Does not include mold geometry or machine plasticizing.
Inline melt sensor / special measurement system	Dynamic melt temp monitoring or validation.	Can reduce air-shot measurement error.	Cost, installation, durability, and calibration must be justified.

7.3 Non-destructive testing for internal flow-line severity

NDT can assess certain internal consequences of poor flow, but it cannot universally grade "flow-line severity" for all plastics. ASNT describes NDT as inspection methods that evaluate materials without causing harm, including visual, radiographic, ultrasonic, infrared, electromagnetic, and other methods [S23]. In injection molded plastics, the most useful methods depend on material opacity, wall thickness, filler content, defect type, and required resolution.

NDT method	Useful for molded plastics	Limitations
Visual / optical	Surface flow lines, splay, burns, gate blush, weld lines, gloss.	Surface only unless transparent part.
Polarized light	Residual stress/orientation in transparent amorphous materials such as PC or PMMA.	Requires transparent or prepared samples; qualitative unless calibrated.
X-ray / micro-CT	Voids, inclusions, knit-line voiding, short glass clumps, metal contamination, internal geometry.	Cost, resolution limits, cycle time; orientation and weld strength not directly measured.
Ultrasonic testing	Voids, delamination, large internal flaws in suitable geometries.	Coupling, attenuation, filler scattering, and thin sections are challenging.
Infrared thermography	Mold thermal imbalance, hot spots, cooling flow issues, part ejection temperature patterns.	Measures surface thermal response; emissivity and timing matter.
DSC / FTIR / MFR lab checks	Material ID, crystallinity, degradation, contamination screening.	Mostly destructive or sample-consuming; not direct NDT of every part.

8. Preventive Measures Before and After the DOE

8.1 Material selection and handling

1. Select resin based on flow length, wall thickness, gate design, mechanical performance, regulatory requirements, and supplier processing window. Do not use flow grade alone to overcome poor tooling.

2. For PP, choose the lowest flow grade that fills within acceptable pressure and quality constraints when toughness matters; processing guides caution that lower-flow resins generally retain better toughness than unnecessarily high-flow grades [S13].

3. For ABS and PC/ABS, control moisture, shear, residence, venting, and color stability before raising melt temperature to chase cosmetics.

4. For PC, PA, PBT, PET, and other moisture-sensitive materials, verify drying with dew point, time, temperature, and moisture testing when applicable.

5. For filled materials, account for screw wear, fiber breakage, gate shear, anisotropic shrinkage, and higher pressure demand.

6. Control regrind percentage, grinder screen, fines, dust, contamination, and heat history. In cold runner tools, runner regrind can become a moving material factor.

8.2 Process optimization from the DOE

The DOE tells the team which variables actually move actual melt temperature and which variables mainly change pressure, appearance, dimensions, or cycle. Use the output to set a window, not a single fragile number. For cold runner molds, the DOE often drives changes to barrel profile, screw rpm, back pressure, fill rate, transfer position, mold water temperature, and regrind control. For hot runner molds, the DOE must also drive manifold/tip setpoints, valve timing, startup purge, interruption limits, and cavity-by-cavity monitoring.

Optimization question	How DOE answers it	Action if result is unfavorable
Is actual melt inside resin window?	Measured melt temp replicates at steady cycle.	Adjust barrel profile, screw rpm, back pressure, residence, nozzle/hot runner zones; verify again.
Is process pressure-limited?	Transfer/peak pressure and velocity response across fill-rate points.	Increase melt within safe window, improve gate/runner, reduce flow length, raise mold temp, or move to larger press.
What fill rate is robust?	Viscosity curve low-slope region plus quality checks.	Use multi-stage fill if one speed cannot satisfy gate, vent, and cosmetic constraints.
Does mold temp mask melt temp?	Temperature DOE with mold temp as factor or block.	Stabilize utilities, repair cooling, set mold-temp alarms.
Is hot runner balanced?	Cavity weights, short shots, cavity pressure, tip PV/SV/output %, gate appearance.	Balance tips, correct wiring/thermocouple fit, repair heater, adjust valve timing, inspect gate area cooling.
Is material degrading?	Color/odor/black speck/MFR/FTIR/DSC trends and residence study.	Lower actual melt/residence, improve purge, repair dead spot, change screw/nozzle/hot runner.

8.3 Mold design improvements

- Gate size should support the selected fill-rate window without excessive shear rate, blush, burn risk, or freeze-off. Gate thickness, land length, and gate type are as important as nominal width.
- Gate location should avoid weak welds in high-load areas, excessive flow length, race-tracking, and late air traps. Simulation plus short shots should verify the pattern.
- Runner design should be balanced by flow and shear history, not only by equal length. Cold runners should minimize pressure loss while controlling cycle time and regrind.
- Hot runner manifolds should be designed for thermal balance, material residence, clean purging, valve-gate access, and independent control where the application risk justifies it.
- Wall thickness should be as uniform as product function allows. Thick-to-thin transitions create hesitation, freeze layers, and localized shear/temperature history shifts.
- Venting must be maintained. Raising melt temperature or speed to defeat trapped gas usually increases burns and degradation rather than solving the root cause.

8.4 Machine calibration and maintenance

- Verify screw/barrel condition, check-ring leakage, heater-band performance, thermocouple seating, pressure transducer calibration, velocity linearity, and clamp parallelism.
- Record screw recovery time and cushion variation as routine process health metrics. Drift in recovery or cushion often precedes melt-quality instability.
- For hot runner molds, treat controller health, cable condition, connector condition, thermocouple type, heater resistance, and zone mapping as part of process validation.

- Do not compensate for a worn screw or leaking check ring by increasing temperature until the mechanical root cause has been assessed.

9. Corrective Actions

9.1 In-process adjustments

Problem found	Cold runner action	Hot runner action	Validation check
Actual melt too low	Increase correct barrel zones, screw rpm, or back pressure; check nozzle/sprue heat loss; raise mold temp if freeze-off dominates.	Check barrel/nozzle first, then manifold/drop/tip zones; inspect cold tip or controller output limit.	Repeat actual melt readings and compare pressure/fill/part data.
Actual melt too high	Lower barrel profile, screw rpm, back pressure, or residence; check shot-size ratio and cycle interruptions.	Lower overheated zones; check stuck relay, thermocouple reversal, dead spots, excessive hot-soak.	Inspect color, odor, black specks, MFR shift, pressure drop.
Pressure high but melt verified OK	Run pressure-loss ladder; inspect gates/runners/vents/cooling.	Check cavity-specific short shots, tip temperature, valve timing, gate obstruction.	Pressure reserve and cavity balance.
Blush / burns	Reduce gate shear, improve venting, use staged fill, adjust melt only after gas/shear confirmed.	Balance hot tips, adjust valve timing, improve venting, review gate design.	Visual grade plus pressure curve and short-shot pattern.
Flash after raising temp	Return to verified melt; correct pack/transfer/clamp/vent/parting-line issue.	Check hot tip overheat and pack sequence; avoid global temperature increase.	Dimensional and flash audit by cavity.
Weld-line weakness	Raise mold temp/melt temp within limits, improve venting, alter fill pattern/gate, strengthen design.	Use sequential valve-gate timing, balance tips, adjust mold/gate temperature.	Mechanical test, microscopy if required.

9.2 Post-processing solutions and their limits

Secondary operations can hide or mitigate cosmetic defects, but they rarely correct the underlying process failure. Polishing can reduce visible flow marks only if the defect is surface-level and the geometry allows it. Painting, pad printing, plating, or coating can hide gloss variation but may introduce adhesion risk if the material is degraded, contaminated, or under high residual stress. Annealing can reduce stress in some materials but may change dimensions. Welding, staking, or assembly operations may fail if weld-line strength is compromised. For regulated or safety components, post-processing is not a substitute for validated molding robustness.

Secondary operation	Can help	Cannot solve
Polishing / buffing	Minor surface haze, flow-line visibility on non-textured surfaces.	Internal weld weakness, degradation, voids, dimensional instability.
Painting / coating	Cosmetic color/gloss mismatch.	Poor substrate integrity, splay voids, contaminated surface without prep validation.
Annealing	Residual stress in selected amorphous/crystalline materials when validated.	Bad fill pattern, poor weld line, burns, gate blush, incorrect material.
Sorting	Short-term containment for visible defects.	Root cause; hidden mechanical defects may escape.
Regrind reuse	Material recovery for cold runner scrap when controlled.	Degraded or contaminated material quality; excess regrind can destabilize DOE assumptions.

9.3 Redesign strategies

- Modify part geometry to remove abrupt thick-to-thin transitions, long thin flow restrictions, uncontrolled race-tracking, and weld lines in high-stress regions.
- Increase gate size or change gate type when shear rate and pressure loss are the limiting factors. Validate vestige, drool, gate seal, and appearance.
- Relocate gates or add sequential gating to control flow-front collision and weld-line location.
- Improve venting and vacuum assist where air traps are the constraint. Do not raise melt temperature to push through trapped air.
- For cold runner molds, redesign runner balance, runner diameter, cold slug wells, and sprue geometry to reduce pressure loss and thermal loss.
- For hot runner molds, redesign manifold balance, drop diameter, tip style, valve pin/gate fit, thermal insulation, and independent zone control where needed.

10. Industry Best Practices and Standards Alignment

Best practice is to document and transfer plastic variables, not merely machine variables. Beaumont/Plastics Technology explicitly notes that actual melt temperature should be documented as the temperature of plastic coming out of the nozzle rather than only barrel settings [S3]. Scientific molding practice uses fill-only studies, viscosity curves, pressure-loss studies, cavity balance, gate seal, cooling studies, and process capability to build a transferable process. Societies and standards do not typically publish a single "melt temperature verification DOE standard," so the defensible approach is to combine material standards, dimensional standards, scientific molding methods, and internal validation requirements.

Source / standard	Relevant use
ASTM D3835 / ISO 11443-type rheometry	Rheological properties at processing-relevant temperature and shear rate. Useful for material characterization and simulation inputs [S2].
ASTM D1238 / ISO 1133-type MFR/MVR	Incoming quality-control flow test; useful but not a full processing viscosity model [S10].
ASTM D3418 / ISO 11357-type DSC	Thermal transitions, melting/crystallization, crystallinity support, contamination/material confirmation [S11].
ISO 20457	Molded part tolerances and acceptance conditions; note that ISO says surface imperfections such as sink marks, flow structures, roughness, and joint lines are not addressed [S12].
SPE / scientific molding literature	Methodology for fill studies, viscosity curves, gate seal, cooling, pressure loss, and process documentation.
Machine OEM tools	ENGEL iQ weight control uses injection pressure curve comparison to compensate viscosity/material quantity shifts [S20]; ARBURG ALS and similar MES tools support data capture and traceability [S21].
Cavity-pressure systems	Kistler ComoNeo and similar systems support optimization, monitoring, and documentation using cavity pressure and mold temperatures [S22].
Hot runner OEM guidance	Husky and Mold-Masters guidance emphasize thermal control, hot-runner balance, gate quality, and maintenance/safety practices [S17, S18, S19].

10.1 Shop-floor best-practice rules

1. Never state that melt temperature was verified unless the actual plastic temperature was measured or directly monitored with a documented method.

2. Never use a viscosity curve taken at one actual melt temperature to justify a process running at another actual melt temperature without verification.

3. Never average multi-cavity hot-runner data until worst-cavity and worst-drop behavior has been reviewed.

4. Never solve venting with melt temperature until venting has been inspected and short-shot evidence supports the conclusion.

5. Never copy a recipe between presses without matching plastic variables: actual melt temperature, fill time, pressure at transfer, cushion, part weight, mold temperature, and cavity balance.

6. Never let a DOE become an uncontrolled startup trial. Stabilize, randomize or block, replicate, and document.

11. Emerging Technologies and Research

The next generation of melt-temperature verification will use more direct sensing, better machine data, connected auxiliary equipment, and AI-assisted process optimization. This does not eliminate the need for process fundamentals. It raises the standard for data discipline.

Technology	How it helps	Current constraint
IoT-enabled machine and auxiliary monitoring	Captures barrel zones, hot runner zones, water temperature/flow, dryer dew point, screw recovery, pressure curves, robot timing.	Data without context becomes noise; sensors must be calibrated and mapped.
Cavity pressure and temperature monitoring	Provides cycle-by-cycle evidence of fill, pack, gate seal, viscosity shifts, and thermal conditions.	Requires sensor placement, maintenance, and trained interpretation.
AI / machine learning melt-temperature estimation	Research shows ML can estimate melt temperature from machine energy and process data when direct measurement is difficult [S9].	Model validity is machine/material-specific; it needs labeled data and change-control discipline.
Adaptive process control	Systems such as ENGEL iQ weight control adjust process based on pressure-curve deviations and viscosity/material changes [S20].	Not a substitute for initial DOE, hot-runner balance, or material validation.
Digital twins / CAE material characterization	Combines shear viscosity, PVT, thermal data, geometry, and process settings for pre-production risk reduction [S6, S7].	Must be validated with short shots, actual pressures, and actual melt temperature.
Advanced materials: nano-additives, self-healing polymers, high-fill compounds	May improve performance or change flow/thermal behavior.	Often more sensitive to dispersion, shear, residence, gate design, and characterization accuracy.

12. Troubleshooting Guide and Flowchart

The following reaction plan is designed for production support. It deliberately separates measurement error, material error, machine error, cold-runner feed error, hot-runner balance error, and mold vent/cooling error. Use it before making broad temperature increases.

Step	Question	Decision / action
1	Is the cell at steady state?	If not, stabilize cycle, mold water, dryer, hot runner zones, and robot timing. Discard startup data.
2	Is the measurement method verified?	If no, calibrate probe, define purge method, take replicates, and solve false-low/false-high technique.

3	Is actual melt outside resin window?	If yes, adjust barrel/screw/back pressure/residence/hot runner zones within supplier limits. Re-measure.
4	Is pressure high with actual melt in window?	Run pressure-loss/short-shot study. Look for gate, runner, venting, cooling, or machine pressure-limit issue.
5	Is the tool hot runner?	If yes, map every zone/drop/cavity. Check thermocouples, heaters, output %, valve timing, and gate-area cooling.
6	Is the tool cold runner?	If yes, separate runner and part data; check sprue, runner size, cold slug, gate freeze, and regrind stability.
7	Does viscosity curve show stable plateau?	If no, check material, gate shear, pressure limit, temperature stability, and fill-only method.
8	Do defects persist after thermal and rheology verification?	Move to mold design, venting, cooling, material, and part geometry corrective actions.
9	Was change validated?	Run capability lot, update setup sheet, alarms, maintenance plan, and change-control documentation.

12.1 Common pitfalls

Pitfall	Why it causes false determination	Prevention
Measuring purge skin	Cooled skin reads low; processor raises temperature unnecessarily.	Probe core immediately using defined method and replicate readings.
Changing barrel and fill speed together	Cannot separate temperature effect from shear-rate effect.	Use DOE matrix or change one factor at a time for verification.
Running viscosity curve while pressure-limited	Commanded speed is not achieved; curve is invalid.	Verify velocity response and pressure reserve at each point.
Ignoring hot runner zones	Nozzle melt is acceptable but one tip is cold/hot.	Zone map, cavity short shots, cavity weights, cavity pressure.
Ignoring mold temperature drift	Pressure/viscosity changes may be cooling changes.	Measure water supply/return, flow, mold surface, and thermal equilibrium.
Using MFR as full rheology	MFR is one-point empirical data and may not reflect processing shear.	Use capillary rheology or validated on-machine curve for process decisions.
Averaging multi-cavity results	Worst cavity can fail while average looks acceptable.	Track cavity-specific weights, dimensions, pressure, and visual grade.
Skipping interruption study in hot runner	Hot soak degradation occurs only after downtime.	Validate restart purge, max idle time, and zone setback policy.

13. Shop-Floor Checklists and Data Templates

13.1 Pre-DOE readiness checklist

- ☐ Resin grade, lot, color, regrind percentage, and drying condition approved.
- ☐ Machine PM status acceptable; check-ring leakage, screw recovery, and cushion repeatability verified.
- ☐ Mold PM status acceptable; vents, gates, cooling, ejection, and parting line inspected.
- ☐ Hot runner zone map verified, if applicable; thermocouples and heaters checked; controller alarms active.
- ☐ Chiller/thermolator supply, return, flow, and setpoint stable.
- ☐ Temperature measurement probe calibrated and method documented.
- ☐ DOE matrix approved by processing, tooling, quality, and maintenance stakeholders.
- ☐ Safety plan for hot purge, high-temperature resin, hot runner work, and lockout/tagout defined.
- ☐ Acceptance criteria defined before data collection begins.

13.2 Example data collection template

Run	Factor levels	Actual melt T reps	Fill time	Transfer pressure	Peak pressure	Part weight by cavity	Visual grade / notes
1	Baseline	___ / ___ / ___	___ s	___	___	C1___ C2___ C3___ C4___	___
2	Low barrel / nominal rpm / nominal BP	___ / ___ / ___	___ s	___	___	C1___ C2___ C3___ C4___	___
3	High barrel / nominal rpm / nominal BP	___ / ___ / ___	___ s	___	___	C1___ C2___ C3___ C4___	___
4	Nominal barrel / high rpm / nominal BP	___ / ___ / ___	___ s	___	___	C1___ C2___ C3___ C4___	___
5	Nominal barrel / nominal rpm / high BP	___ / ___ / ___	___ s	___	___	C1___ C2___ C3___ C4___	___

13.3 DOE sign-off template

Function	Required review question	Approval / notes
Process Engineering	Does the selected process use verified actual melt temperature and validated fill-rate window?	
Tooling	Were gates, vents, cooling, runner/hot-runner conditions, and cavity balance reviewed?	
Maintenance	Are screw/barrel/nozzle/heaters/controllers/thermolators in qualified condition?	
Quality	Are CTQs, capability data, inspection method, and change-control requirements satisfied?	
EHS	Are purge, hot material, hot runner, and lockout/tagout hazards controlled?	
Production	Can technicians run the setup sheet and reaction plan repeatably on the shift?	

14. References and Further Reading

The following sources were used to ground this document. Verify current revision status before citing a standard in customer, regulatory, or contractual documentation.

[S1] RJG, "What Is Viscosity and How to Monitor It in Injection Molding Processes." https://rjginc.com/have-a-molding-problem-the-answer-might-be-viscosity/

[S2] ASTM D3835, Standard Test Method for Determination of Properties of Polymeric Materials by Means of a Capillary Rheometer. https://store.astm.org/d3835-16.html

[S3] Hoffman & Beaumont, "A New Look at Evaluating Fill Times For Injection Molding," Plastics Technology / Beaumont Technologies. https://www.beaumontinc.com/files/plasticstechnology_noads3.pdf

[S4] Plastics Technology, "Injection Molding: Fast, Simple, Low-Cost Method Unveils True Melt Temperature." https://www.ptonline.com/articles/injection-molding-fast-simple-low-cost-method-unveils-true-melt-temperature

[S5] PlasticsToday, "Injection Molding: The Mystery of the Melt Temperature." https://www.plasticstoday.com/injection-molding/injection-molding-the-mystery-of-the-melt-temperature

[S6] Moldex3D, "Material Characterization." https://www.moldex3d.com/products/service/material-characterization/

[S7] Autodesk, Moldflow Material Testing Overview.
https://help.autodesk.com/sfdcarticles/attachments/MPL_Material_Testing_Overview.pdf

[S8] Autodesk Moldflow Help, Process Settings Wizard - Fill + Pack Settings.
https://help.autodesk.com/view/MFIA/2024/ENU/?guid=MoldflowInsight_CLC_Analyses_analysis_sequences_Fill_analysis_Fill_Pack_analysis_Process_Settings_Wizard_dialog_15_html

[S9] Jeon, Gim & Rhee, "Melt Temperature Estimation by Machine Learning Model Based on Energy Flow in Injection Molding," Polymers, 2022. https://www.mdpi.com/2073-4360/14/24/5548

[S10] ASTM D1238, Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer.
https://store.astm.org/d1238-20.html

[S11] ASTM D3418, Standard Test Method for Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by DSC. https://webstore.ansi.org/standards/astm/astmd341821

[S12] ISO 20457:2018, Plastics moulded parts - Tolerances and acceptance conditions.
https://www.iso.org/standard/68097.html

[S13] INEOS Polypropylene Processing Guide. https://www.ineos.com/globalassets/ineos-group/businesses/ineos-olefins-and-polymers-usa/products/technical-information--patents/ineos_polypropylene_processing_guide.pdf

[S14] SABIC Injection Molding Processing Guide. https://www.controlplastics.com/technical-resources/SABIC-PLA-751_Injection_Molding.pdf

[S15] DuPont Delrin Acetal Resin Molding Guide. https://dupont.materialdatacenter.com/links/processing/Delrin.pdf

[S16] BASF Ultramid Product Brochure / Processing Guidance. https://www.basf.com/dam/jcr%3A5bfdf5e3-5ec8-33bb-a216-9cd7c258a772/basf/www/cn/documents/en/chinaplas/Ultramid_brochure.pdf

[S17] Mold-Masters Hot Runner User Manual and Troubleshooting Guide.
https://www.moldmasters.com/sites/default/files/2024-08/ENG%20SPT%20UM.pdf

[S18] Husky Hot Runner Systems. https://www.husky.co/en/solutions/hot-runners/

[S19] Husky, "Hot Runners Play a Key Role in Optimizing System Balance." https://www.husky.co/en/resources/blog/hot-runners-play-a-key-role-in-optimizing-system-balance/

[S20] ENGEL, "Compensate for fluctuations in plastic viscosity."
https://www.engelglobal.com/en/us/digital-solutions/digital-solutions-injection-molding-manufacturing-process/optimize-plastic-viscosity

[S21] ARBURG Host Computer System ALS. https://www.arburg.com/en/us/product-world/digital-products/als-host-computer-system/

[S22] Kistler ComoNeo process monitoring and control system. https://www.kistler.com/US/en/cp/process-monitoring-and-control-system-comoneo-5887b/P0000382

[S23] American Society for Nondestructive Testing, NDT Methods.
https://www.asnt.org/what-is-nondestructive-testing/methods

[S24] Eng-Tips forum thread, "Melt Temperature." https://www.eng-tips.com/threads/melt-temperature.119054/

[S25] Reddit r/InjectionMolding discussion, "Melt temperature."
https://www.reddit.com/r/InjectionMolding/comments/1o1cnj3/melt_temperature/

[S26] Moldex3D, Molding Simulation: Theory and Practice.
https://ch.moldex3d.com/wp-content/uploads/2018/06/Contents-of-Molding-Simulation-Theory-and-Practice.pdf

[S27] Protolabs, "A Guide to the Scientific Injection Molding Process." https://www.protolabs.com/resources/guides-and-trend-reports/scientific-molding-helps-ensure-repeatable-part-function/

[S28] BASF, Injection-Molding Problems in Engineering Thermoplastics - Causes and Solutions.
https://download.basf.com/p1/8a8082587fd4b608017fd6631d5a24b1/en/Injection-Molding_Problems_in_Engineering_Thermoplastics_-_Causes_and_Solutions_Brochure_German.pdf?view=

[S29] Plasticstoday, “Scientific Molding, Part 1: Filling.” https://www.plasticstoday.com/blow-molding/scientific-molding-part-1-filling

[S30] ASNT, Radiographic, ultrasonic, infrared, visual, and related NDT method descriptions. https://www.asnt.org/what-is-nondestructive-testing/methods

14.1 Video, webinar, and forum resources for technician training

Resource type	**Recommended use**	**Cautions**
Machine/OEM videos and webinars	Show controller functions, hot runner balance concepts, cavity pressure monitoring, and adaptive control.	Do not treat marketing claims as validation data for your mold.
SPE, RJG, Beaumont, and Plastics Technology training	Good foundation for scientific molding, viscosity curves, pressure-loss studies, and process documentation.	Training methods must be adapted to each resin, mold, and quality system.
Eng-Tips and practitioner forums	Useful for seeing real troubleshooting questions and field symptoms.	Anecdotal; verify through controlled measurement before acting.
Autodesk Moldflow / Moldex3D tutorials	Useful for simulation setup, interpreting shear/pressure/temperature results, and comparing short shots.	Simulation must be validated against actual process data.

Appendix A - Safety and EHS Requirements

Hot polymer purge can cause severe burns. Some resins release hazardous decomposition products when overheated. Acetal, PVC, fluoropolymers, flame-retarded materials, and unknown contamination require resin-specific safety review. Hot runner work can expose personnel to high voltage, stored heat, pressurized plastic, and pinch points. Follow the machine, mold, hot runner, resin supplier, and site EHS procedures. Lockout/tagout is required for maintenance tasks. The DOE must never require an operator to catch purge by hand, bypass guards, defeat interlocks, or reach into the mold area during unsafe machine states.

Appendix B - Glossary

Term	Definition
Actual melt temperature	Physical temperature of the molten polymer at a defined location and cycle state, measured or monitored by a defined method.
Barrel setpoint	Controller target for a heater zone. It is not the same thing as actual melt temperature.
Relative viscosity index	Shop-floor comparative metric often calculated from fill time and pressure or pressure-curve area. It is not laboratory viscosity.
Fill-only shot	Shot transferred before pack/hold influence, often used in viscosity curves and short-shot studies.
V/P transfer	Velocity-to-pressure transfer point between fill and pack/hold.
Hot runner tip delta	Difference between a tip/drop temperature and a reference manifold/nozzle/melt temperature.
Pressure loss study	Sequential diagnostic to locate pressure demand from nozzle through runner, gate, and cavity.
Cavity balance	Degree to which cavities fill, pack, cool, and produce weights/dimensions consistently.
Thermal equilibrium	Stable machine/mold/runner state where repeated cycles produce repeatable thermal and pressure data.